MAILAM Engineering College

Mailam, Villupuram(Dt), Pin: 604 304
(Approved by AICTE, New Delhi,Permanently Affiliated toAnna University Chennai,
Accredited by NBA,NAAC with 'A' Grade and TATA Consultancy Services)

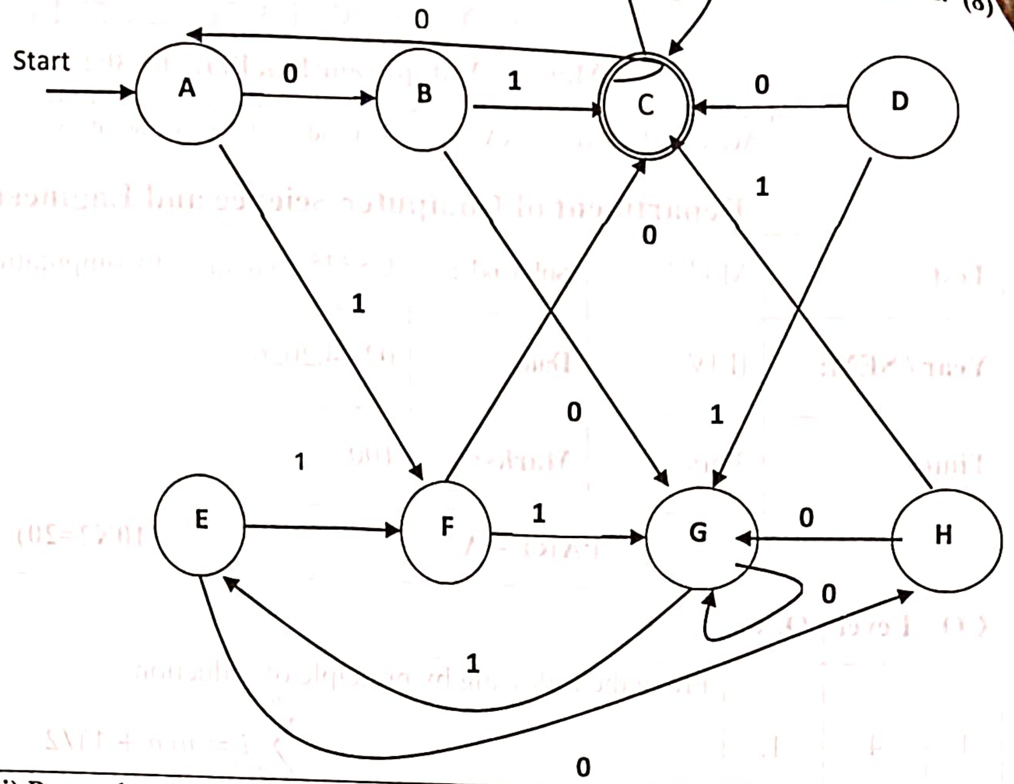
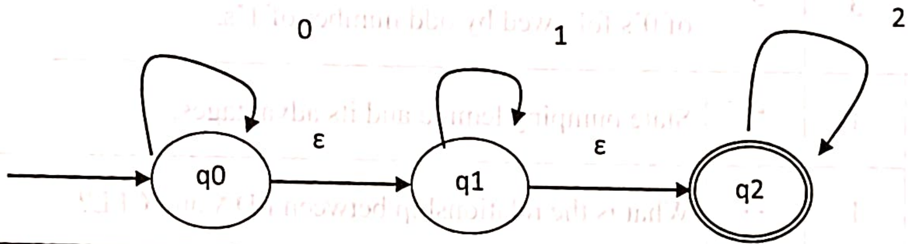
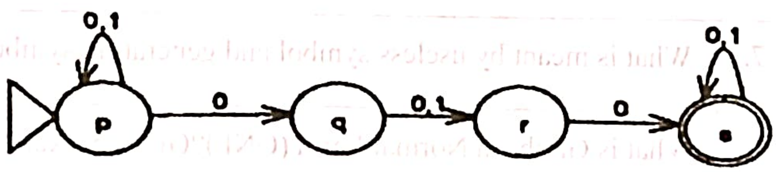
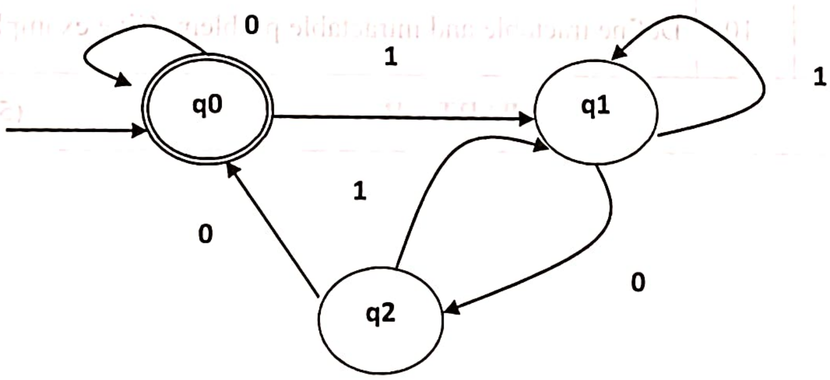
Department of Computer Science and Engineering

Test	Model	**Sub/Code:**	CS3452/Theory of Computation
Year / SEM:	II/IV	**Date:**	02.04.2026
Time:	3 hrs	**Marks:**	100

PART – A **(10X2=20)**

CO	Level	Q.No	
1	4	1.	Prove the following by principle of Induction. $$\sum_{i=0}^{n} i = n(n+1)/2$$
1	4	2.	Design a DFA for the language recognized by the automaton being L= { ab^n : n ≥ 0 }
2	3	3.	Write a regular expression for the set of string over {0,1}, string contains even number of 0's followed by odd number of 1's.
2	1	4.	State pumping lemma and its advantages.
3	1	5.	What is the relationship between PDA and CFL?
3	3	6.	Is it NPDA is more powerful than DPDA? Justify.
4	2	7.	What is meant by useless symbol and generating symbol? Give example.
4	2	8.	What is Greibach Normal Form (GNF)?Give an Example.
5	3	9.	When do you say a problem is decidable? Give example.
5	2	10.	Define tractable and intractable problem. Give example.

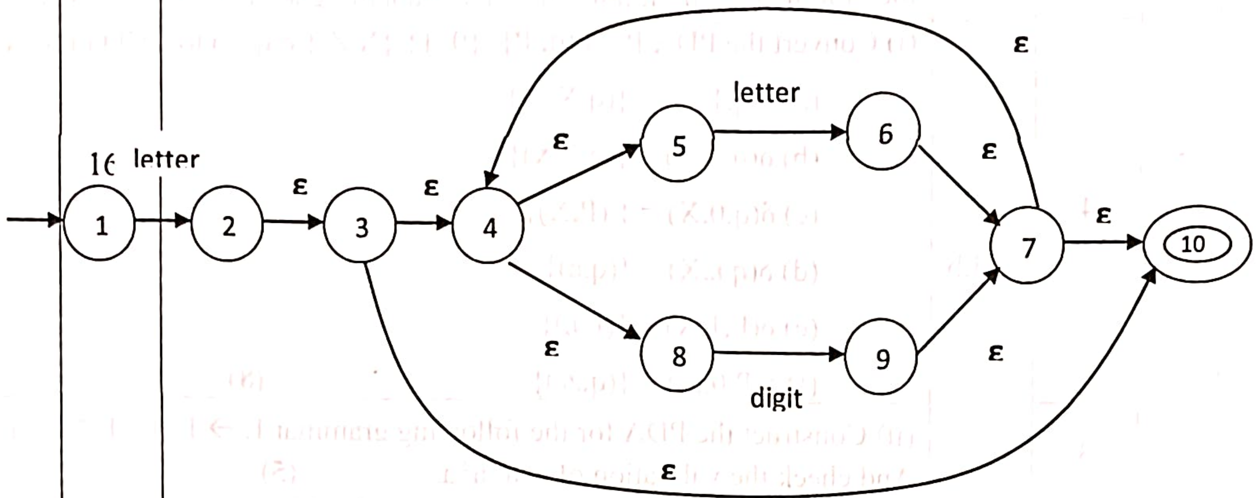
PART – B **(5X13=65)**

1	4	11.a	Construct a minimal state DFA and find the regular expression for the DFA. (8) 
	4		(ii) Prove that for every L accepted by an ε-NFA, then there exists L which is accepted by NFA without ε transitions. (5)
	4	11.b	(i) Convert the following NFA with ε to DFA directly. (7) 
	4		(ii) Find the equivalent DFA for the following NFA. (6) 
2	4	12.a	(i)Find the regular expression corresponding to the Deterministic finite automata (DFA) given below. (8) 

	4		(ii) Construct a FA that can be derived from the following Regular Expression 0*(01)(0/111)* **(5)**
	2	12.b	(i) Explain in detail about closure properties of Regular languages (RL). **(7)**
	3		(ii) Prove the following statement with justification. "The language L= { $a^i\, b^j\, c^i$ / i,j>0} is not regular". **(6)**
3	4	13.a	(i) Examine whether the language, L = { $a^{2n}\, b^p\, c^{2n}$/ n, p >0} can be designed using pushdown automation. Justify your answer. **(7)**
	2		(ii)Define ambiguous grammar. Consider the grammar S→aS | aSbS | Є. Show that the grammar is ambiguous and create unambiguous grammar. **(6)**
	4	13.b	(i) Convert the PDA P = ({q, P}, {0, 1},{X,Z_0},δ,q,z_0) to a CFG if δ is given by, (a) δ(q,1,z_0) = {(q,Xz_0)} (b) δ(q,1,X) = {(q,XX)} (c) δ(q,0,X) = { (P,X)} (d) δ(q,ε,X) = {(q,ε)} (e) δ(P,1,X) = {(P,ε)} (f) δ(P,0,z_0) = {(q,z_0)} **(8)**
	3		(ii) Construct the PDA for the following grammar E → E+E | E * E | a And check the validation of a*a+a*a. **(5)**
4	4	14.a	(i) What is the purpose of normalization? Construct the CNF and GNF for the following grammar and explain the steps. **(8)** S→a|Aa|B $S \rightarrow aBB \mid$ Є $B \rightarrow Aa \mid b$
	3		(ii) Prove that L= {a^p | P is divisible by 4 } is not context free language. **(5)**
	4	14.b	(i)Define Turing Machine. Design a Turing Machine to compute f(m,n) = m−n , if m≥n = 0 , if m<n. Also list out the programming techniques of Turing machine. **(8)**
	3		(ii) Explain in detail about Closure properties of Context Free Languages. **(5)**
5	3	15.a	(i)Give short notes on Recursive and Recursive Enumerable Languages. **(7)** (1) Prove that Ld is not recursively enumerable. (2) Prove that Lu is recursively enumerable.
	4		(ii) Let Σ = {0, 1} , Let A and B be the list of string defined as

	List A	List B
i	wi	xi
1	10	101
2	011	11
3	101	011

Find the instance of MPCP. **(6)**

	3	15.b	(i) With proper examples, explain P and NP complete problems. Show that 3-CNF SAT is NP complete. (8)
	3		(ii) Prove that If there is a reduction from P 1 to P 2 then (1) If P 1 is undecidable, then so is P 2 (2) If P 1 is non-RE, then so is P. (5)
			PART – C (1X15=15)
3	3	16	(i) Define DPDA. and Design a DPDA to accept the language L = {WCW^R / W∈{0,1}} . (7)
1	4		(ii)Consider the following ε-NFA for an identifier. Consider the ε-closure of each state and find it's equivalent DFA. **(8)** 
4	4	16.b	(i) Define halting problem. Explain in detail about universal Turing Machine with a proper example. **(8)**
5	4		(ii) Explain the philosophy behind Travelling salesman problem (TSP). Analyze the computational complexity for the same. Show how the decision version of the TSP belongs to the class of NP-complete problem. Also explain about kruskal's Algorithm. (7)

CO1: Construct automata theory using Finite Automata
CO2: Write regular expressions for any pattern
CO3: Design context free grammar and Pushdown Automata
CO4: Design Turing machine for computational functions
CO5: Differentiate between decidable and undecidable problems

K1: Remember **K2:** Understand **K3:** Apply **K4:** Analyze **K5:** Evaluate **K6:** Create

Subject In-charge **Batch Co-ordinator** **HOD/CSE** **Principal**